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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

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5539296252

MATHEMATICS (SYLLABUS D) **4024/11**

Paper 1 **October/November 2011**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

For Examiner's Use

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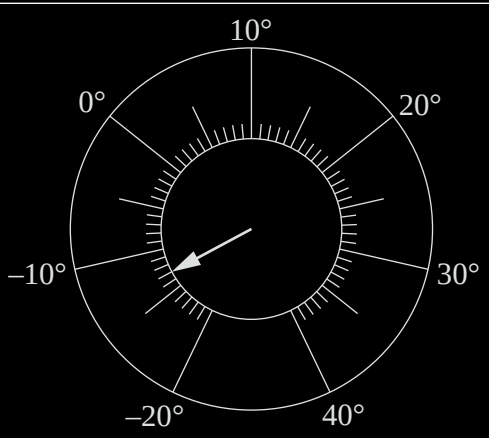
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Answer ..... °C [1]

(b) When the temperature increases from –20 °C to 40 °C, the pointer turns through an angle of 300°.

Calculate the angle through which the pointer turns when there is a 7 °C rise in temperature.

Answer ..... [1]

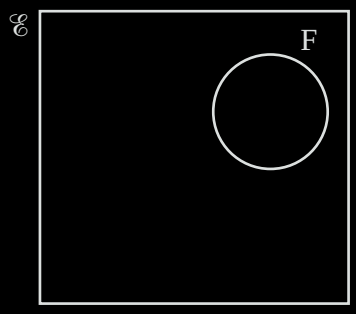
(c) On one particular day, the temperature at 1 a.m. was 4 °C.
By 6 a.m. it had fallen by 9 °C.

Calculate the temperature at 6 a.m.

Answer ..... °C [1]

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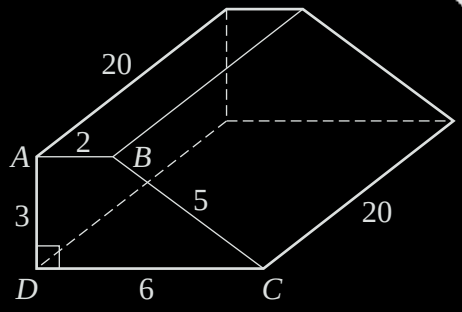


(a) Draw and label the two sets P and S to complete the Venn diagram. [2]

(b) Write down a possible element y such that y is an even number and $y \in (F \cup S)'$.

Answer $y =$ ..... [1]

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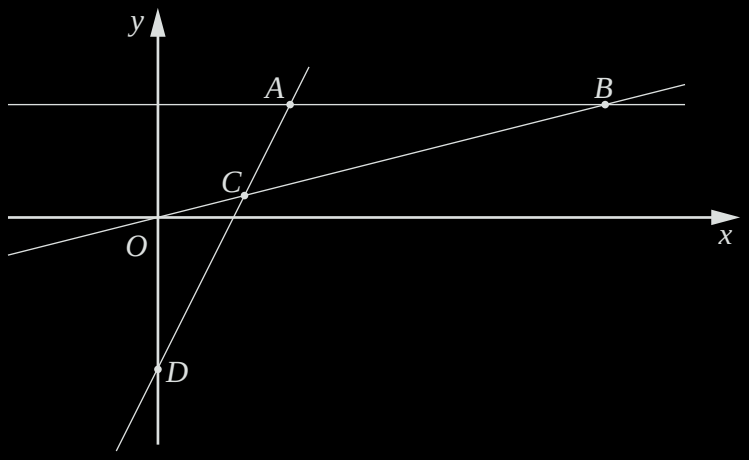
(a) Calculate the area of trapezium *ABCD*.

Answer ..... cm^2 [1]

(b) Calculate the **total** surface area of the prism.

Answer ..... cm^2 [2]

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In the diagram, B is the point (8, 2).
The equation of the line AB is $y = 2$ and the equation of the line AC is $2x - y = 3$.
BC produced passes through the origin.

(a) AC produced intersects the y-axis at D.

Find the coordinates of D.

Answer (..... , ...) [1]

(b) The region **inside** triangle ABC is defined by three inequalities.
One of these is $y < 2$.

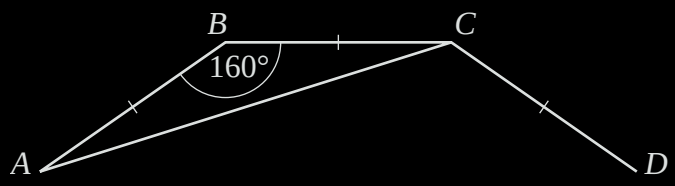
Find the other two inequalities.

Answer .....

..... [2]

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The diagram shows three sides, *AB*, *BC* and *CD*, of this polygon.

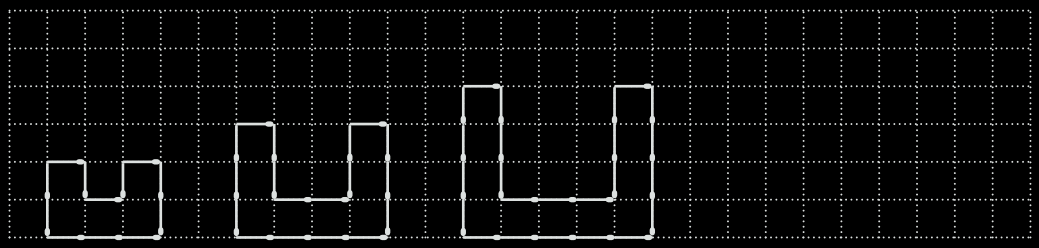
(i) Calculate $B\hat{A}C$.

Answer ..... [1]

(ii) Calculate $A\hat{C}D$.

Answer ..... [1]

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Shape 1 Shape 2 Shape 3 Shape 4

(a) Draw Shape 4. [1]

(b) The table shows the numbers of matchsticks used to make Shapes 1 and 2.

Shape	1	2	3	4
Number of matchsticks	12	18		

Complete the table for Shapes 3 and 4. [1]

(c) Find an expression, in terms of n, for the number of matchsticks used to make Shape n.

Answer ..... [1]

(d) Explain why there is not a shape that is made of 100 matchsticks.

Answer .....

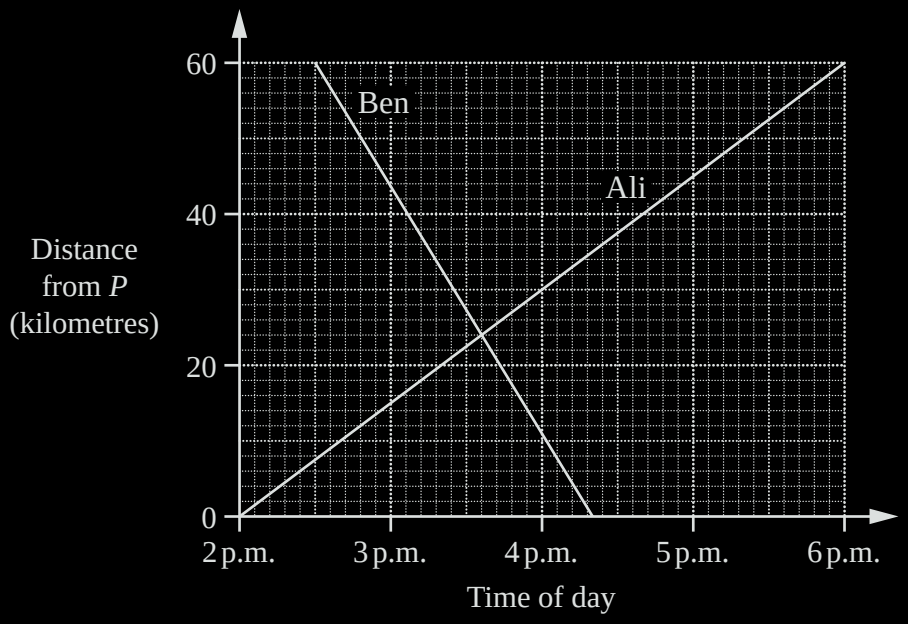
.....

..... [1]

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Ali and Ben each made a journey between two towns, P and Q, that are 60 km apart.
These two journeys are shown on the travel graph.

(a) Calculate Ali's speed.

Answer ..... km / h [1]

(b) Find the number of minutes after 3 p.m. that Ali and Ben passed each other.

Answer ..... [1]

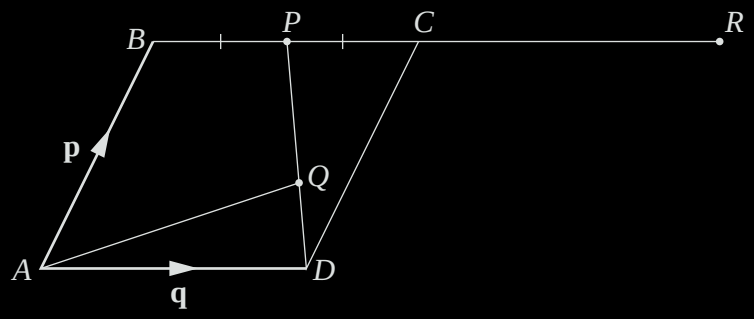
(c) Find how far Ben had travelled when he met Ali.

Answer ..... km [1]

(d) Chris left P at 3 p.m. and travelled to Q at a speed of 30 km / h.

On the diagram, draw the graph that represents Chris's journey. [1]

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In the diagram, $ABCD$ is a parallelogram.
P is the midpoint of BC.
$DQ : QP = 1 : 2$.
$\overrightarrow{AB} = \mathbf{p}$ and $\overrightarrow{AD} = \mathbf{q}$.

(a) Express $\overrightarrow{DP}$ in terms of **p** and **q**.

Answer ..... [1]

(b) Express $\overrightarrow{DQ}$ in terms of **p** and **q**.

Answer ..... [1]

(c) Express $\overrightarrow{AQ}$ in terms of **p** and **q**, giving your answer in its simplest form.

Answer ..... [1]

(d) R is the point on BC produced such that $\overrightarrow{BR} = k\overrightarrow{BP}$.

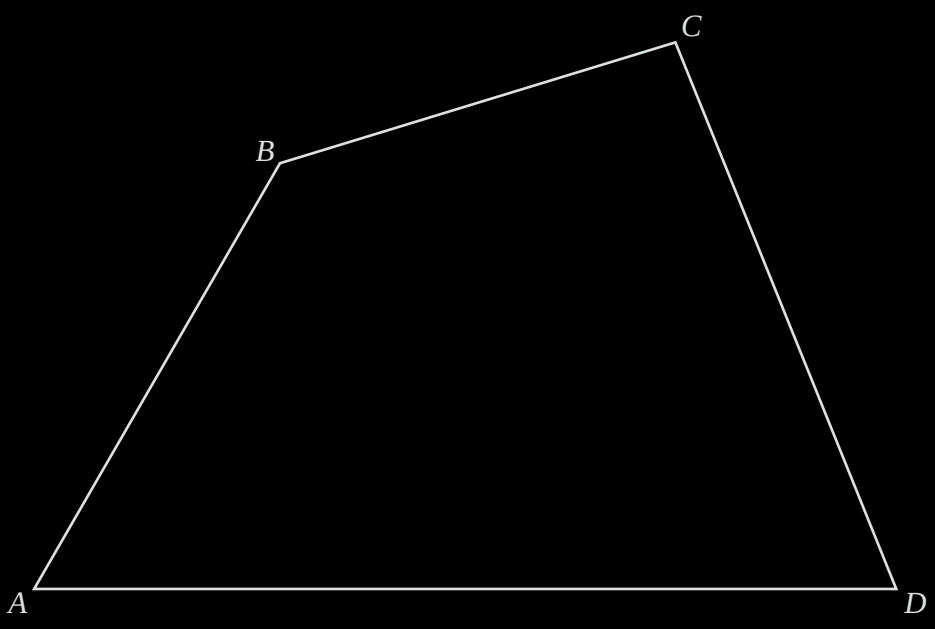
(i) Express $\overrightarrow{AR}$ in terms of **p** and **q** and k.

Answer ..... [1]

(ii) Given that A, Q and R lie on a straight line, find the value of k.

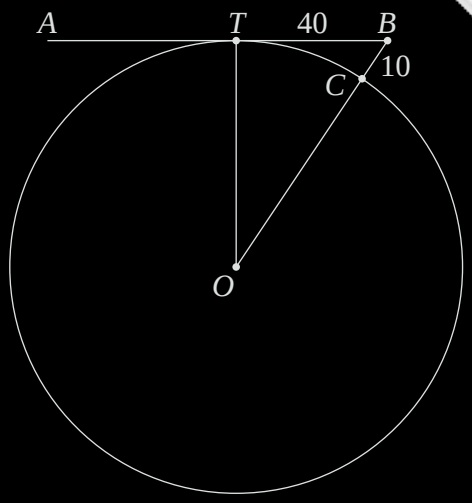
Answer $k =$ ..... [1]

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Question 26 is printed on the following page.

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(a) State, with a reason, the value of $B\hat{T}O$.

Answer $B\hat{T}O$ = ... because .....

..... [1]

(b) Given that $TB = 40$ cm, $CB = 10$ cm, and the radius of the circle is x centimetres, form an equation in x, and hence find the radius of the circle.

Answer ..... cm [4]